Roney Lima do Nascimento

AI Solutions Developer | Applied AI Researcher | PhD Candidate in Pure Mathematics, IME-USP

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PROFILE

Applied AI engineer with sixteen years of Python and production systems across the full large language model lifecycle: fine-tuning with LoRA adapters, embeddings, retrieval-augmented generation, evaluation and deployment. I lead AI implementation end to end, from architecture and governance through to release, monitoring and staff enablement, and I have shipped platforms serving more than fifteen thousand monthly users across thirty countries. My engineering sits on a mathematical foundation: PhD candidate in Pure Mathematics at IME-USP with the qualifying examination passed in August 2026, which is why my work on model behaviour starts from evidence and uncertainty quantification rather than from demonstrations. Author of two books on generative AI. Delivered a keynote online at ICAILY 2026. Claude Certified Architect.

SELECTED PRODUCTION SYSTEMS

- **HELIX**, applied AI research and orchestration engine. More than 121,000 lines of code and 3,800 automated tests, currently at version 6. Multi-model routing, evaluation harness and long-horizon agent memory.
- **AI Teachers** (ai-teachers.pro). Production platform serving more than 15,000 monthly users across 30 countries. Full stack, from data model and API design to deployment and observability.
- **Paulean AI**. Retrieval-augmented assistant over structured and unstructured institutional data, 1,100 users. Vector store, semantic retrieval pipeline and access-controlled document ingestion.
- **IBPREDICTIONS**. Machine learning prediction service using a stacking ensemble with hyperparameter optimisation and conformal prediction intervals, so every prediction ships with a calibrated uncertainty bound.
- **Financial automation agent** for a corporate client. Automated income-statement extraction, reconciliation and reporting from heterogeneous sources, replacing a manual monthly close process. Packaged desktop deployment with an audit trail.
- **Two Mistral-7B LoRA adapters** published on Hugging Face for British English prompt optimisation, trained on custom datasets.
- **Automated assessment engine**. OCR plus multimodal LLM marking with rubric-aligned evaluation and formative feedback at batch scale. Peer-reviewed paper under review at Springer.

EXPERIENCE

AI Solutions Developer and Consultant, independent practice — Jan 2023 to present

- Design and deliver production AI for companies and schools: retrieval-augmented assistants, agentic workflows, machine learning prediction pipelines and document-processing systems.
- Lead AI implementation as a programme, not a pilot: opportunity assessment, architecture, data governance and LGPD posture, build, evaluation, rollout and training.
- Run AI enablement programmes for teams of more than eighty professionals, and teacher-training programmes for schools in Brazil and abroad.

AI Solution Developer, whole school, St Paul's School, São Paulo — Feb 2025 to Jun 2026

- Owned the institution's AI delivery: integrations with iSAMS, the learning platform and SharePoint, automated reporting and scheduling, RAG assistants and predictive dashboards for early intervention.
- Ran pilots with impact evaluation, and embedded privacy, ethics and safeguarding requirements into every solution rather than bolting them on afterwards.

Teacher of Applied Generative AI and of IBDP Mathematics HL, Colégio São Luís — May 2026 to present

- Lead the applied generative AI elective, taking frontier model capability into classroom practice with assessment integrity and ethical guardrails.
- Teach the two-year IB Diploma Programme Higher Level mathematics syllabus, plus differential calculus and computational thinking in the Ensino Médio.

Teacher of Mathematics, IBDP Higher Level, KS3 and IGCSE, St Paul's School, São Paulo — Dec 2023 to May 2026

Earlier teaching, Colégio Dante Alighieri, Escola Internacional de Alphaville, Colégio Objetivo and others — 2004 to 2024

Fifteen years of mathematics teaching across public, private and international curricula, including research supervision in the Cientista Aprendiz programme and volunteer teaching in community pre-university courses.

TECHNICAL STACK

Languages	Python (since 2010), TypeScript, JavaScript, SQL, Node.js, POSIX shell
AI and ML	PyTorch, Hugging Face Transformers, PEFT and LoRA, LangChain, scikit-learn, XGBoost, LightGBM, Optuna, conformal prediction, ChromaDB, FAISS, OpenAI and Anthropic APIs
Engineering	REST API design, OAuth, Flask, React, Git and GitHub, CI/CD, automated testing at scale, Docker
Cloud and data	Google Cloud Platform, AWS, Firebase, Supabase and PostgreSQL, SharePoint Framework, Linux VPS administration

EDUCATION

PhD, Pure Mathematics, Universidade de São Paulo (IME-USP). Flat ternary cyclotomic polynomials, supervised by Prof. Raul Antonio Ferraz. Qualifying examination passed August 2026. CAPES fellow. 2024 to 2028

MBA, Artificial Intelligence, Data Science and Big Data for Business, Ibmec (450 hours) 2025

MBA, Leadership, Innovation and Technology, Ibmec 2024

MSc, Mathematics, Universidade de São Paulo 2014 to 2017

BSc, Mathematics, Universidade de São Paulo 2010 to 2013

CERTIFICATIONS

Claude Certified Architect, Foundations (Anthropic, 2026) · IBM Generative AI Engineering · Google Generative AI Leader (valid to Dec 2028) · Applied Generative AI, Johns Hopkins (100 hours) · GitHub Foundations (valid to May 2028) · Microsoft Innovative Educator Expert 2026 · AI Product Management · Generative AI, Vanderbilt and UPenn Wharton

PUBLICATIONS, SPEAKING AND PEER REVIEW

Books. *Generative AI for Teachers: A Practical Guide to Educational Technology* (2025) and *Inteligência Artificial Generativa para Professores* (Editora Dialética, 2026). The Portuguese book was launched at Livraria da Vila on 9 October 2026.

Speaking. Delivered a keynote online at ICAILY 2026 and a presentation at DataMinitech. MiningTech South America 2026, São Paulo. Summit de Inteligência Artificial do Brasil 2026, Joinville.

Teacher development. Delivered four AI training sessions at Colégio São Luís during September and October 2026, with teachers from early years, primary and secondary education.

Peer review. Education and Information Technologies (Springer). Programme committee and emergency reviewer, International Conference on Educational Data Mining 2026.

Writing on AI evaluation and governance. 18 original articles published in 2026 as of 14 October, including in Folha de S.Paulo, The Fulcrum, HEPI, Educational Leadership (ISTE and ASCD), EUobserver, GovInsider, Business Day (South Africa), eSchool News and Nexo Políticas Públicas.

LANGUAGES

Portuguese (native) · English (full professional) · Spanish (professional working)